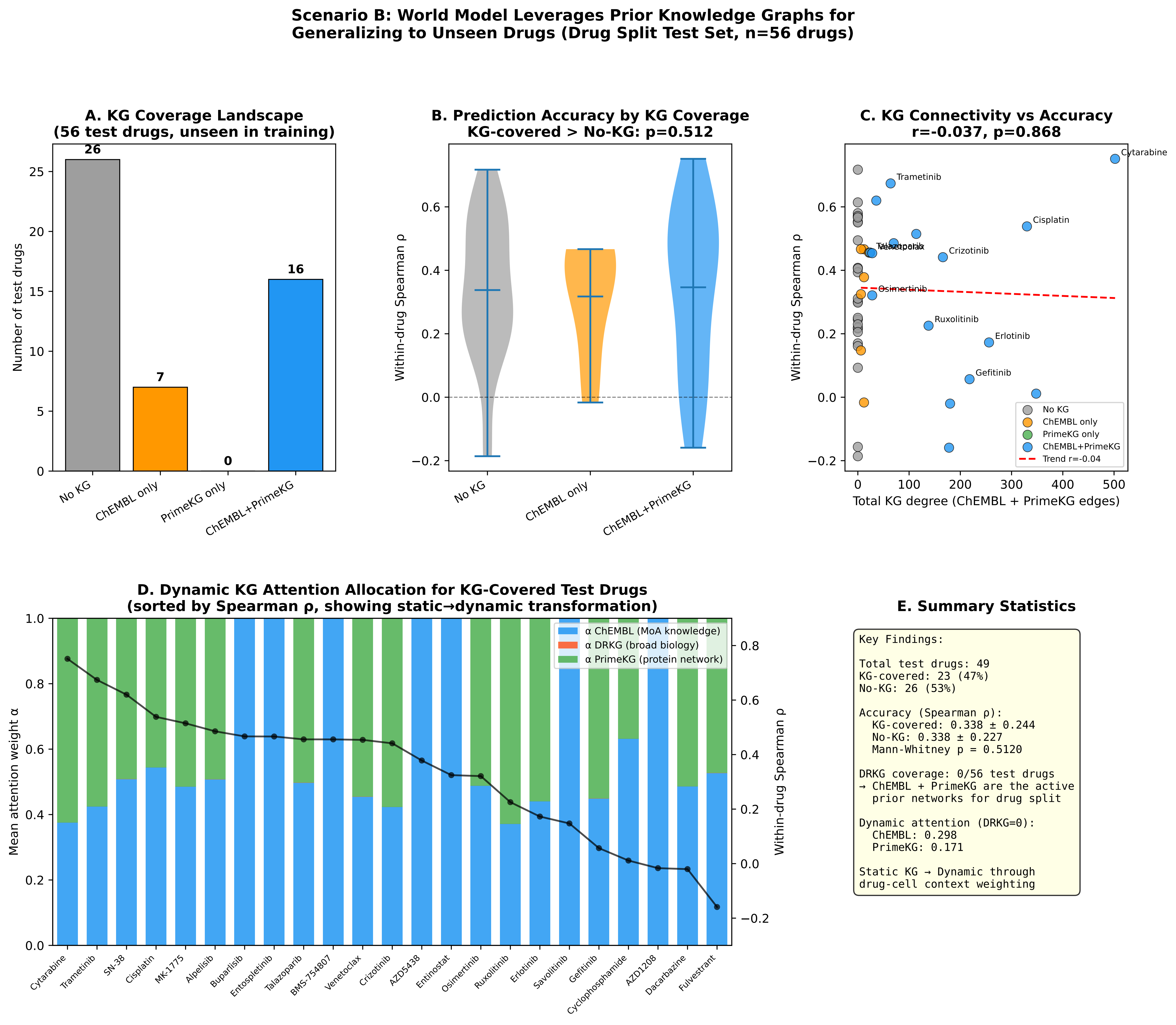
Scenario B: World Model Leverages Prior Knowledge Graphs for
Generalizing to Unseen Drugs (Drug Split Test Set, n=56 drugs)
A. KG Coverage Landscape
(56 test drugs, unseen in training)
Number of test drugs
26
7
0
16
No KG
ChEMBL only
PrimeKG only
ChEMBL+PrimeKG
B. Prediction Accuracy by KG Coverage
KG-covered > No-KG: p=0.512
Within-drug Spearman ρ
No KG
ChEMBL only
ChEMBL+PrimeKG
C. KG Connectivity vs Accuracy
r=-0.037, p=0.868
Within-drug Spearman ρ
Total KG degree (ChEMBL + PrimeKG edges)
Cytarabine
Trametinib
Cisplatin
Crizotinib
Osimertinib
Ruxolitinib
Erlotinib
Gefitinib
No KG
ChEMBL only
PrimeKG only
ChEMBL+PrimeKG
Trend r=-0.04
D. Dynamic KG Attention Allocation for KG-Covered Test Drugs
(sorted by Spearman ρ, showing static→dynamic transformation)
Mean attention weight α
Within-drug Spearman ρ
α ChEMBL (MoA knowledge)
α DRKG (broad biology)
α PrimeKG (protein network)
Cytarabine
Trametinib
SN-38
Cisplatin
MK-1775
Alpelisib
Buparlisib
Entospletinib
Talazoparib
BMS-754807
Venetoclax
Crizotinib
AZD5438
Entinostat
Osimertinib
Ruxolitinib
Erlotinib
Savolitinib
Gefitinib
Cyclophosphamide
AZD1208
Dacarbazine
Fulvestrant
E. Summary Statistics
Key Findings:
Total test drugs: 49
KG-covered: 23 (47%)
No-KG: 26 (53%)
Accuracy (Spearman ρ):
KG-covered: 0.338 ± 0.244
No-KG: 0.338 ± 0.227
Mann-Whitney p = 0.5120
DRKG coverage: 0/56 test drugs
→ ChEMBL + PrimeKG are the active
prior networks for drug split
Dynamic attention (DRKG=0):
ChEMBL: 0.298
PrimeKG: 0.171
Static KG → Dynamic through
drug-cell context weighting